

In the middle of a thin rod of length L , closed on both ends, a small ampoule containing the mass M of a dye (color). At $t = 0$, the ampoule breaks, and the dye diffuses in the rod. The ampoule is assumed to be 'pointlike'. Formulate the mathematical model for the dye concentration for $t > 0$ and solve the problem.

Solution:

Diffusion,

$$\begin{aligned}\frac{\partial u}{\partial t} - D \frac{\partial^2 u}{\partial x^2} &= 0, & 0 < x < L, t > 0 \\ \frac{\partial u(0, t)}{\partial x} &= \frac{\partial u(L, t)}{\partial x} = 0, & t > 0 \\ u(x, 0) &= M \delta_{\frac{L}{2}}, & 0 < x < L\end{aligned}$$

We have homogenous Neumann boundary conditions, which should give us a solution in the form of a cosine series expansion.

$$u(x, t) = \sum u_k(t) \cos\left(\frac{k\pi x}{L}\right)$$

Differentiating termwise and putting into the differential equation gives

$$\begin{aligned}\sum (u'_k(t) + \alpha u_k(t)) \cos\left(\frac{k\pi x}{L}\right) &= 0, \quad \alpha = \frac{Dk^2\pi^2}{L^2} \\ \downarrow \\ u'_k(t) + \alpha u_k(t) &= 0 \\ \downarrow \\ u_k(t) &= c_k e^{-\alpha t}\end{aligned}$$

We can also expand $u(x, 0) = M \delta_{\frac{L}{2}}$ in a cosine series, to get the coefficients, $u_k(0) = c_k$

$$u(x, 0) = M \delta_{\frac{L}{2}} = \sum a_k \cos\left(\frac{k\pi x}{L}\right)$$

$$c_0 = \frac{1}{L} \int_0^L M \delta_{\frac{L}{2}} dx = \frac{M}{L}$$

$$c_k = \frac{2}{L} \int_0^L M \delta_{\frac{L}{2}} \cos\left(\frac{k\pi x}{L}\right) dx = \frac{2M}{L} \cos\left(\frac{k\pi}{2}\right) = , \quad k = 1, 2, 3...$$

This gives the full solution

$$u(x, t) = \frac{M}{L} \left(1 + 2 \sum \cos\left(\frac{k\pi}{2}\right) e^{-\alpha t} \cos\left(\frac{k\pi x}{L}\right) \right)$$

□